I got the power

Principled Bayesian dynamic borrowing with power prior methods

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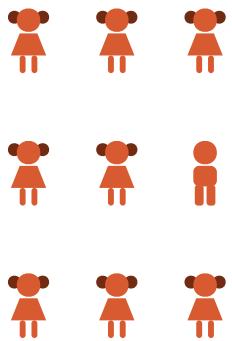
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Motivation

Paediatric RCT

PLUTO, n = 92



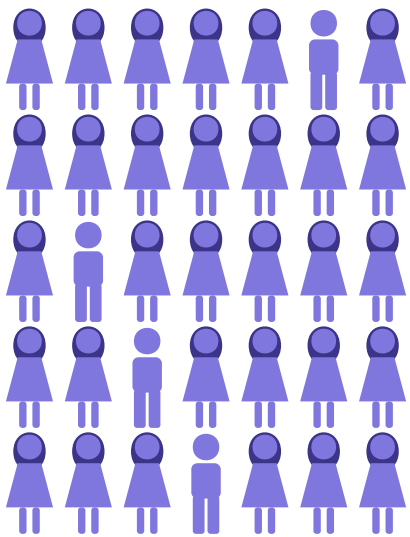
few, hard to recruit

borrow?

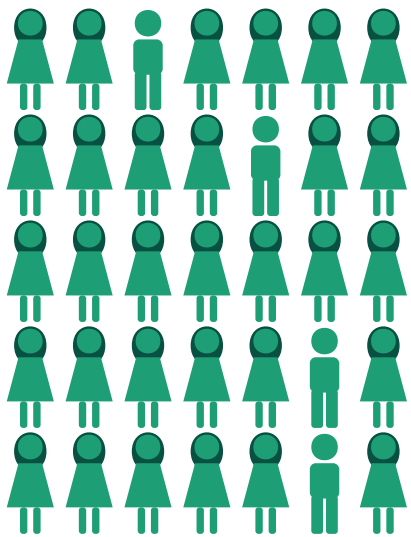
how much?

Adult trials

same drug, same endpoint



BLISS-52, n = 548



BLISS-76, n = 577

Setup

Let

- $\mathbf{D_0} = (D_0^{(1)}, \ldots, D_0^{(n_0)}) \sim P_0 \equiv P_{\theta_0}$ be the **historical** data;
- $\mathbf{D} = (D^{(1)}, \ldots D^{(n)}) \sim P_1 \equiv P_{\theta_1}$ be the **current** data.
- Model is well-specified

Power prior

$$\pi_{\text{PP}}^{(\eta)}(\theta|D_0) := \frac{L(D_0|\theta)^\eta \pi_0(\theta)}{c_0(\eta)},$$

Normalised power prior

$$\pi_{\text{NPP}}(\theta, \eta \mid D_0) := \pi_{\text{PP}}^{(\eta)}(\theta \mid D_0)\pi_A(\eta) \propto \frac{L(D_0 \mid \theta)^\eta}{c_0(\eta)}\pi_0(\theta)\pi_A(\eta).$$

Main objects

$$
\begin{aligned}
p_{\text{NPP}}(\theta, \eta \mid D_0, D) &:= L(D \mid \theta)\pi_{\text{NPP}}(\theta, \eta \mid D_0), \\
&\propto \frac{L(D \mid \theta)L(D_0 \mid \theta)^{\eta}}{c_0(\eta)}\pi_0(\theta)\pi_A(\eta), \\
&\propto p(\theta \mid D)L(D_0 \mid \theta)^{\eta}\frac{\pi_A(\eta)}{c_0(\eta)}.
\end{aligned}
$$

- $p^T_{\text{NPP}}(\theta \mid D_0, D) \equiv$ marginal effect; of central interest
- $p^A_{\text{NPP}}(\eta \mid D_0, D) \equiv$ tells us something about compatibility

Roadmap

"Coherent" borrowing

How do we make $\eta \to 0$ when $P_0 \neq P_1$ **and** $\eta \to 1$ when $P_0 = P_1$?

Principled computation

NPP is doubly-intractable; can we compute exactly? Are there **good**, efficient approximations?

Multiple historical data sets

Strange things (can) happen when you have multiple historical data sets.

Principled borrowing

Oliveira et al (2026+), Shen et al. (2026)[1]:

Theorem (NPP shrinks to zero under discrepancy)

If $P_0 \neq P_1$, under suitable conditions,

$$p^A_{\text{NPP}}(\eta \mid D_0, D) \to \delta_{\{0\}}(\eta),$$

a.s. $P^n_{\theta_1} \otimes P^{n_0}_{\theta_0}$.

Unwanted(?) discounting

However, when $P_0 = P_1$ the NPP does **not** concentrate at 1, and this is seen as a problem (Pawel et al. , 2023).

[1] and others.

Why though?

Theorem: inability to fully borrow

if π_A is continuous and positive on $(0, 1]$, then

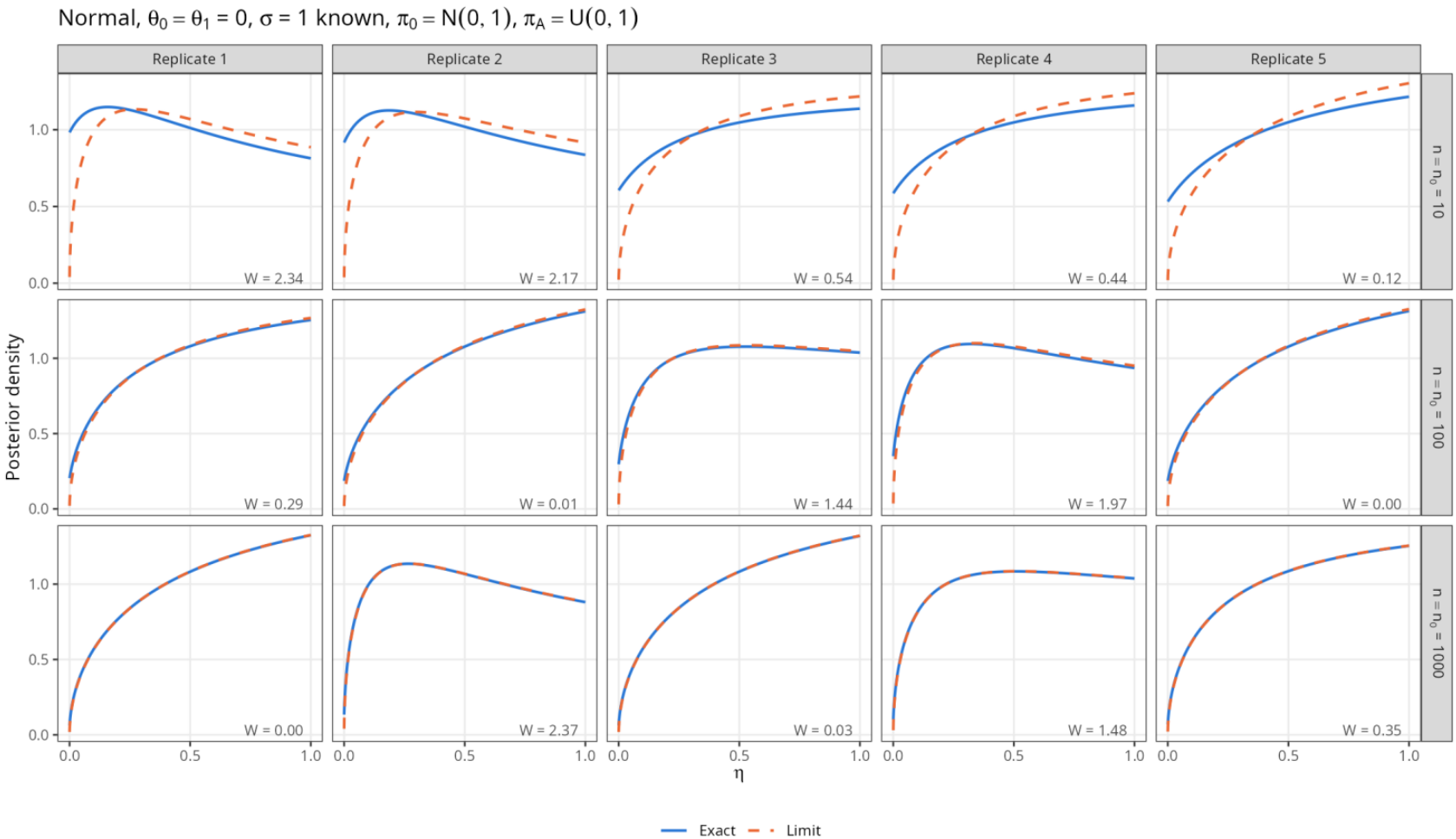
$$d_{\mathrm{TV}}\big(p_{\mathrm{NPP}}(\eta \mid D, D_0),\ g_n(\eta)\big) \xrightarrow{p} 0, \quad \text{whence}$$

$$\begin{aligned} g_n(\eta) &\propto \lambda_r(\eta)^{p/2} e^{-\lambda_r(\eta) W_n/2} \pi_A(\eta), \\ W_n &= (\hat{\theta}_1 - \hat{\theta}_0)^\top (J_1^{-1} + J_0^{-1})^{-1} (\hat{\theta}_1 - \hat{\theta}_0), \\ \lambda_r(\eta) &= \frac{\eta(1+r)}{1+\eta r}. \end{aligned}$$

In particular, the posterior does not converge to a point mass at $\eta = 1$.

Proof idea(s): under regularity assumptions, do a Laplace approximation of the marginal likelihood under compatibility.

Illustration: normal model with known variance



A couple of tentative solutions

Data-adaptive, fixed η

Choose η in a data-dependent manner, but keep it fixed

Information matching power prior (IMPP)

Matching information between the power prior and the regular posterior. This is an adaptation of Holmes & Walker (2017).

Predictive criterion power prior (PCPP)

Making the predictive criterion of Reznik (2026) work on real data.

Information matching power prior (IMPP)

We choose η such that

$$\int_{\mathcal{X}^{k_n}} I_\eta(x) L(x \mid \theta_0) \mathrm{d}x = \int_{\mathcal{X}^{k_n}} I_1(x) L(x \mid \theta_1) \mathrm{d}x,$$

where $k_n = \lfloor (rn)^\nu \rfloor = \lfloor n_0^\nu \rfloor$ for $\nu \in [0, 1]$, $I_\eta(x) = \eta^2 \Delta(x)$, and $\Delta(x)$ is the Fisher relative information divergence of the posterior update from $p(\theta \mid D)$ to $p(\theta \mid D, x)$.

$$\Delta(x) = \int \sum_{j=1}^{d} \left\{ \sum_{i=1}^{k_n} \frac{\partial}{\partial \theta_j} \log L(x_i \mid \theta) \right\}^2 p(\theta \mid D) \mathrm{d}\theta$$

IMPP and its properties

This setup leads to (Braga & Oliveira et al. (2026+)):

Information matching power prior (IMPP)

$$\hat{\eta}_{\text{IM}} = \left(\frac{\int_{\mathcal{X}^{k_n}} \Delta(x) L(x \mid \hat{\theta}_1) \mathrm{d}x}{\int_{\mathcal{X}^{k_n}} \Delta(x) L(x \mid \hat{\theta}_0) \mathrm{d}x} \right)^{1/2}.$$

Fixed variance normal example

Suppose that $y_{0i} \sim \mathcal{N}(\theta_0, \sigma^2)$ with likelihood $L_0(y_{0i} \mid \theta_0, \sigma^2)$, and $y_i \sim \mathcal{N}(\theta_1, \sigma^2)$.

$$\eta = \left(\frac{\frac{v}{\sigma^2} + \frac{1}{\sigma^2}(m - \theta_1)^2 + \frac{1}{k_n}}{\frac{v}{\sigma^2} + \frac{1}{\sigma^2}(m - \theta_0)^2 + \frac{1}{k_n}} \right)^{1/2}.$$

Predictive criterion

Under a parametric model of dimension d, define

$$K_i(\theta, \vartheta) = D_{KL}(p_\theta(\cdot \mid x_{0i}), p_\vartheta(\cdot \mid x_{0i})),$$

historical design points.

Predictive criterion power prior (IMPP)

The theoretical optimum based on the full historical sample is

$$\eta^*_{\text{PC}} = \frac{d}{d + 2\sum_{i=1}^{n_0} K_i(\theta_0, \theta_1)}.$$

We consider using

$$\hat{\eta}_{\text{PC}} = \frac{d}{d + 2\sum_{i=1}^{k_n} K_i(\hat{\theta}_0, \hat{\theta}_1)}.$$

Here, have your "coherence"

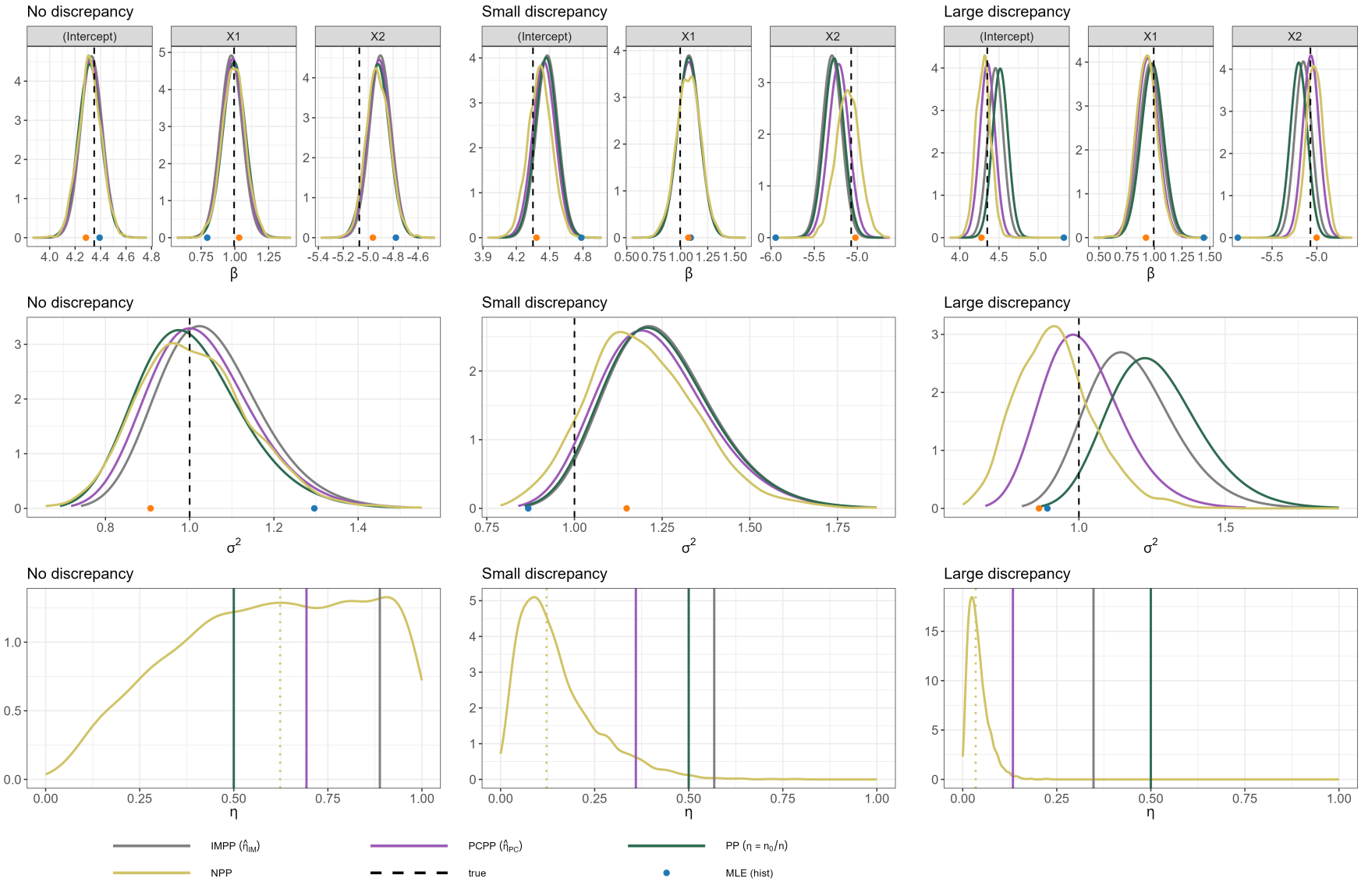
Theorem

Let $D \sim P_{\theta_1}^n$ and $D_0 \sim P_{\theta_0}^{n_0}$ with $r = n_0/n$ a fixed constant. Under appropriate conditions, for a given sequence $\{k_n\}_{n=1}^{\infty} \subset \mathbb{N}$ with $k_n \leq n_0$ we have the following implications:

(i) *If $k_n \to \infty$, then $\hat{\eta}_{\text{IM}} = O_p(k_n^{-1/2})$ whenever $\theta_0 \neq \theta_1$, and, in particular $\hat{\eta}_{\text{IM}} \xrightarrow{p} 0$;*

(ii) *If $k_n/n \to 0$, then $\hat{\eta}_{\text{IM}} = 1 + O_p(k_n/n + n^{-1/2})$ whenever $\theta_0 = \theta_1$, and, in particular, $\hat{\eta}_{\text{IM}} \xrightarrow{p} 1$.*

Results


No discrepancy
Small discrepancy
Large discrepancy
(Intercept)
X1
X2
β
σ²
η
IMPP ($\hat{\eta}_{IM}$)
NPP
MLE (curr)
PCPP ($\hat{\eta}_{PC}$)
true
PP ($\eta = n_0/n$)
MLE (hist)

Variational Normalised Power Prior (VNPP)

- Careful asymptotic analysis: we prove new and general BvM-type results for the PP and the NPP;
- This motivates a MVN variational approximation $\rightarrow$ VNPP

Theorem (Variational approximation under compatibility)

Suppose that $\theta_0 = \theta_1$. Under appropriate conditions, if $(1/n_0) \ll \varepsilon_{n_0} \ll 1$, we have that

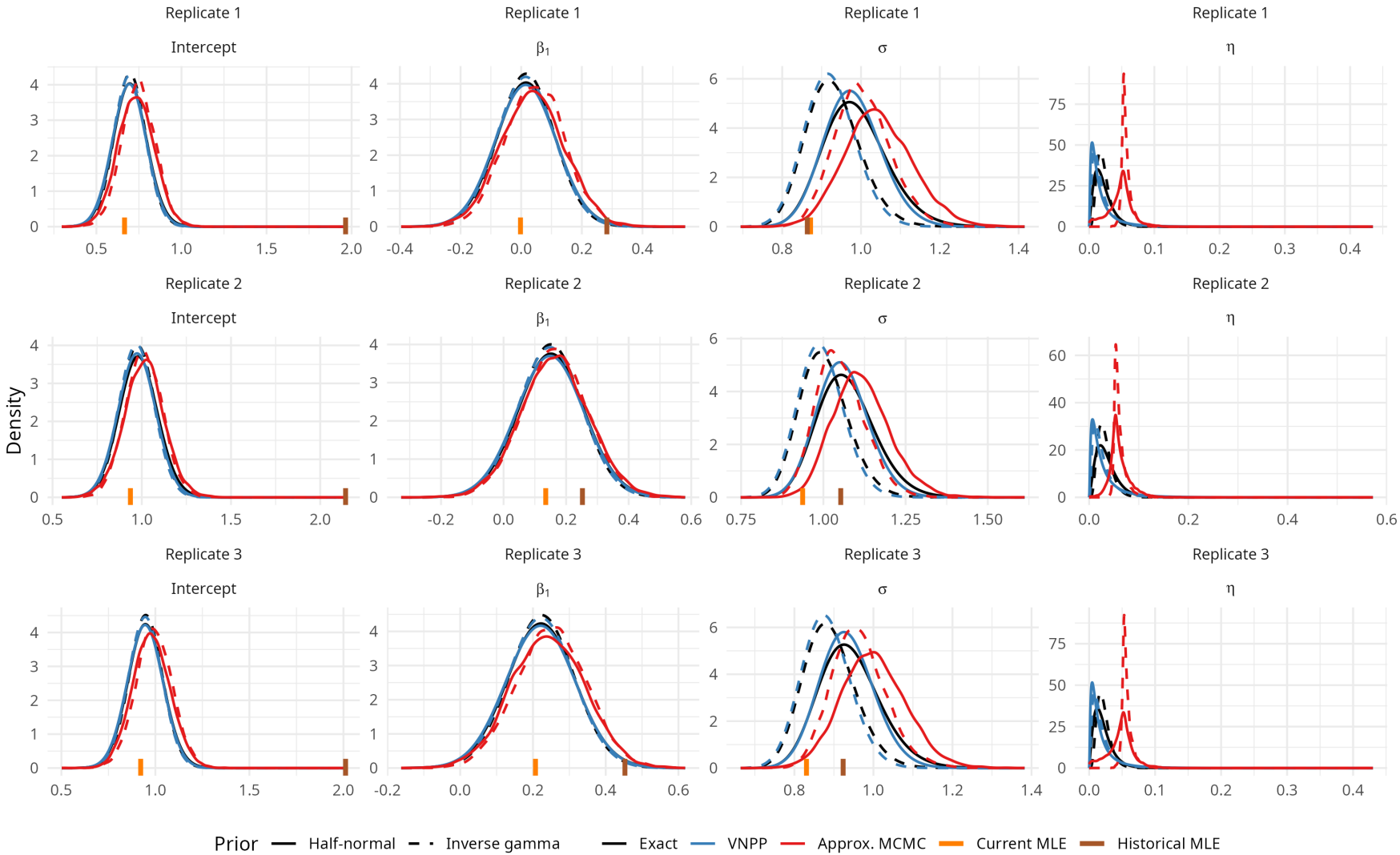
$$\mathbb{P}\left[\int D_\eta(P_\theta, P_{\theta_0})\tilde{p}^{(\eta)}_{\mathrm{PP}}(\mathrm{d}\theta \mid D, D_0) \leq c(r,\eta)\right] \geq 1 - \frac{1}{n_0\varepsilon_{n_0}}h(r).$$

whence

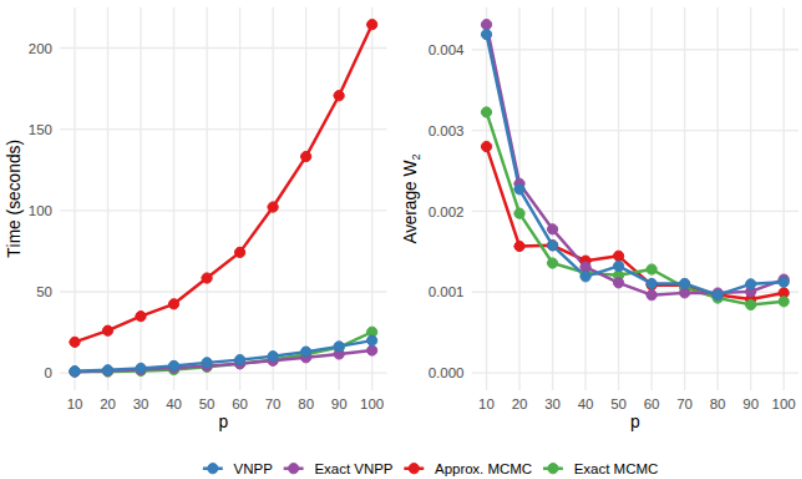
$$c(r,\eta) := 2\left(3 + \frac{1}{r}\right)\frac{1+\eta}{1-\eta}\varepsilon_{n_0},$$

$$h(r) := 1 + \frac{1}{r} + \frac{2}{3 + 1/r}.$$

VNPP works!


Replicate 1
Intercept
β1
σ
η
Replicate 2
Replicate 3
Density
Prior
Half-normal
Inverse gamma
Exact
VNPP
Approx. MCMC
Current MLE
Historical MLE

VNPP works! (2)



Parameter	Method	Bias	MSE	WIS	Coverage (95%)
β_0	Approx. MCMC	-0.001	0.005	0.034	0.966
	Exact MCMC	-0.001	0.005	0.034	0.968
	VNPP	-0.001	0.005	0.033	0.962
	Exact VNPP	-0.001	0.005	0.033	0.962
β_1	Approx. MCMC	-0.005	0.006	0.035	0.958
	Exact MCMC	-0.005	0.006	0.034	0.960
	VNPP	-0.005	0.006	0.034	0.960
	Exact VNPP	-0.005	0.006	0.034	0.960
σ	Approx. MCMC	-0.031	0.003	0.026	0.954
	Exact MCMC	-0.031	0.003	0.026	0.956
	VNPP	-0.027	0.003	0.025	0.938
	Exact VNPP	-0.027	0.003	0.025	0.940

Multiple historical datasets are a menace

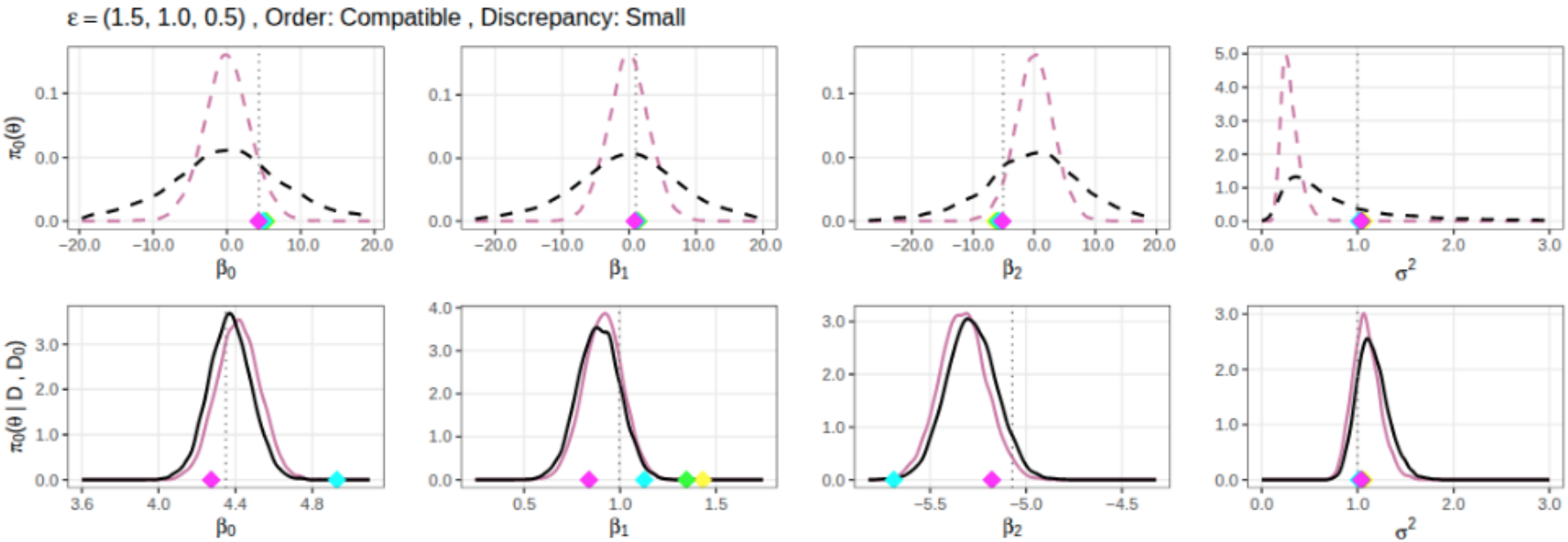
Normalisation matters

If you have data sets $D_{0k}, k = 1, 2, \ldots, K$, how do you actually do things? Jointly or sequentially?

$$\pi_{\text{NPP}}\left(\boldsymbol{\theta}, \boldsymbol{\eta} \mid \boldsymbol{D}_0\right) = \frac{\pi_A(\boldsymbol{\eta})\pi_0\left(\theta\right)\prod_{k=1}^{K} L_k\left(D_{0k} \mid \boldsymbol{\theta}\right)^{\eta_k}}{c_0\left(\boldsymbol{\eta}\right)},$$

$$\pi_{\text{NPP}_{\text{SEQ}}}\left(\boldsymbol{\theta}, \boldsymbol{\eta} \mid \boldsymbol{D}_0\right) = \pi_A(\boldsymbol{\eta}) \prod_{k=1}^{K} p_{\eta_k}\left(\boldsymbol{\theta} \mid D_{0k}\right).$$

Consequences!



Take home

It's possible to get "coherent" borrowing

But you give up on convergence speed

Variational approximations with guarantees

Beats the state-of-the-art in the accuracy-tractability trade-off

Multiple historical data sets

Are the future, but require careful prior elicitation.

Power priors provide semi-automatic learning

But other alternatives and connections (e.g. to hierarchical models) should be investigated – see Shen et al. (2026b) and Chen & Braga et al. (2026+)

THE END